UNIVERSITAS INDONESIA

Chapter 5 Introduction to Valuation – The Time Value of Money

I Putu Sukma Hendrawan, M.S.M, CSA

The learning objectives

What we are about to learn during this chapter

Future value of an investment made today

Present value of cash to be received at a future date

Return on an investment

The **length of time** taken for an investment to reach a desired value

Key terms

Terms	Description
Present Value (PV)	value of a cash flow at an earlier period on a time line → discounting
Future Value (FV)	value of a cash flw at some time in the future → compounding
Annuity	finite series of equal payments at regular interval: Ordinary annuity → at the end of the period Annuity due → at the beginning of the period
Perpetuity	Infinite series of equal payment, annuity that last forever
Principal	Original amount that borrowed or invested
Interest rate	The discount rate, the ocst of capital, the opportunity cost of capital, the required of return. The rate that compensates for the loss of earnings and the loss of purchasing power.
Compounding	The process of accumulating interest on an investment over time to earn more interest
Interest on interest	Interest earned on th reinvestment of previous interest payments
Compound interest	Interest earned on both the initial principal and the interest reinvested from prior periods
Simple interest	Interest earned only o the original principal amount invested

Basic Formula

Future Value (FV)

$$FV = C \times (1 + \boldsymbol{r})^{t}$$

Cash flow at current period

Future Value Interest Factor (FVIF)

Present Value (FV)

$$PV = C \times \frac{1}{(1 + \boldsymbol{r})^{t}}$$

Cash flow at t period of time in the future

Present Value Interest Factor (PVIF)

Number of period

Simple and Compound Interest

10% x beginning amount at year-1

10% x ending amount at previous period

Year	Beginning Amount	Simple Interest	Compound Interest	Total Interest Earned	Ending Amount
1	$100.00	$10	$ 0.00	$10.00	$110.00
2	110.00	10	1.00	11.00	121.00
3	121.00	10	2.10	12.10	133.10
4	133.10	10	3.31	13.31	146.41
5	146.41	10	4.64	14.64	161.05
		Total simple interest $50	Total compound interest $11.05	Total interest $61.05	

UNIVERSITAS INDONESIA

Chapter 6 Discounted Cash Flow Valuation

I Putu Sukma Hendrawan, M.S.M, CSA

The learning objectives

What we are about to learn during this chapter

Future value and present value with **multiple cash flow**

Loan Calculation and the interest rate

Amortization of loan

Interest rate **quotation**

Basic Formula

Future Value (FV) of Multiple Cash Flow

$$FV = \sum_{t=1}^{n} [C \times (1 + \boldsymbol{r})^{t}]$$

Present Value (FV) of Multiple Cash Flow

$$PV = \sum_{t=1}^{n} \left[C \times \frac{1}{(1 + \boldsymbol{r})^{t}} \right]$$

Basic Formula

Future Value (FV) of Annuity

$$FV = C \times \frac{(FVIF - 1)}{r}$$

Future Value Interest Factor Annuity (FVIFA)

Present Value (FV) of Annuity

$$PV = C \times \frac{(1 - PVIF)}{r}$$

Present Value Interest Factor Annuity (PVIFA)

Basic Formula

Annuity Due Value

$$Annuity\ Due = Ordinary\ Annuity\ \times (1 + \mathrm{r})$$

Perpetuity

$$PV = \frac{C}{r}$$

Effective Annual Rate

$$EAR = \left(1 + \frac{APR}{m}\right)^{m} - 1$$

Effective Annual Rate (EAR) The interest rate expressed as if it were compounded once per year.

Annnual Percentage Rate (APR) The interest rate charged per period multiplied by the number of periods per year

Type of Loan

Interest-only loan: periodic interest and end of tenure principle

Amortized loan: periodic interest and principal repayment, fixed principal

Amortized loan with fixed payment: periodic interest and principal repayment, fixed total payment

Year	Beginning Balance	Total Payment	Interest Paid	Principal Paid	Ending Balance
1	$5,000	$1,450	$ 450	$1,000	$4,000
2	4,000	1,360	360	1,000	3,000
3	3,000	1,270	270	1,000	2,000
4	2,000	1,180	180	1,000	1,000
5	1,000	1,090	90	1,000	0
Totals		$6,350	$1,350	$5,000	

Year	Beginning Balance	Total Payment	Interest Paid	Principal Paid	Ending Balance
1	$5,000.00	$1,285.46	$ 450.00	$ 835.46	$4,164.54
2	4,164.54	1,285.46	374.81	910.65	3,253.88
3	3,253.88	1,285.46	292.85	992.61	2,261.27
4	2,261.27	1,285.46	203.51	1,081.95	1,179.32
5	1,179.32	1,285.46	106.14	1,179.32	0.00
Totals		$6,427.30	$1,427.31	$5,000.00	

Thank You

I Putu Sukma Hendrawan, M.S.M, CSA

https://linkedin.com/in/iputusukma
0812-9654-9229
iputusukma@pm.me
https://iputusukma-book.github.io